2 The Earth has a mass M of 5.97×10^{24} kg. A satellite of mass m orbits the Earth in a circular geostationary orbit at a distance d above the Earth's surface.

(a) Show that the radius of the Earth, R, is approximately 6370 km.

[1]

(b) By considering the orbit of the geostationary satellite, show that the distance d above the Earth's suface is 35900 km.

[2]

(c) For the satellite of mass m that is in a circular orbit of radius r around the Earth of mass M, show that its total mechanical energy E can be expressed as $E = -\frac{GMm}{2r}$.

[2]

(d) The satellite has a mass of 490 kg and carries 360 kg of lithium battery pack with energy density of 324 Wh kg^{-1}, of which 65% of its energy is converted to useful energy for orbital manoeuvres.

(i) Show that the amount of useful energy is 2.73×10^8 J.

[1]

(ii) Calculate the maximum orbital radius r_{max} the satellite can reach from its geostationary orbit.

r_{max} = ... m [2]

[Total: 8]